

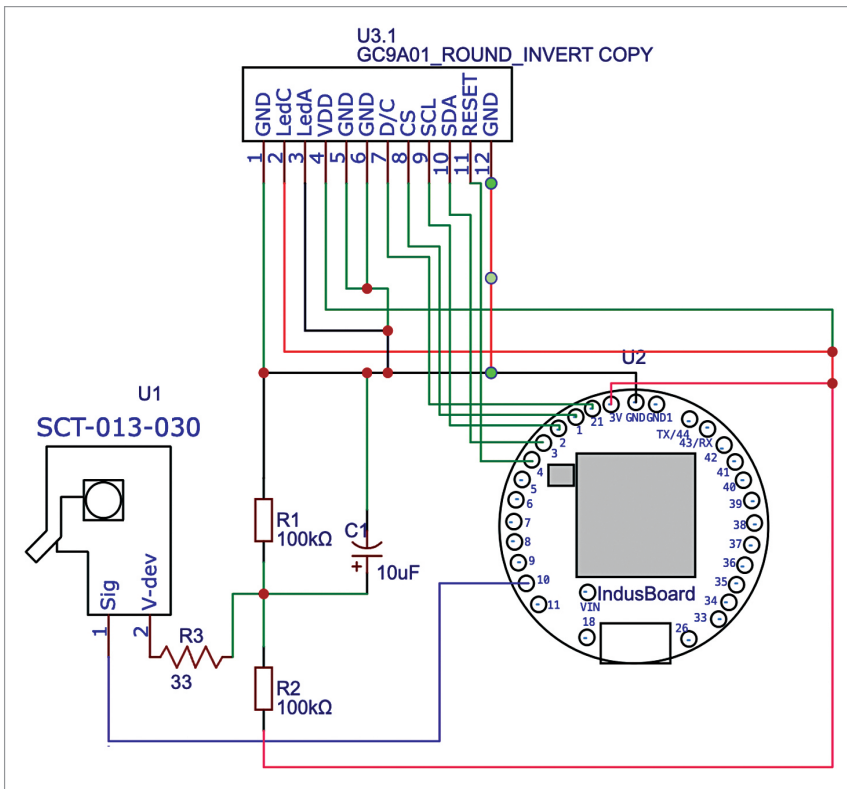


Fig. 3: The IoT meter circuit with GC9A01 round display

Non-invasive and plug-and-play. Eliminates the need to alter existing wiring. You may just attach the sensor to start monitoring.

Compact and portable. Its lightweight design makes it easy to move and use in various locations.

Flexible usage. Easy to plug and unplug, allowing users to measure the electricity usage of different devices by effortlessly switching between them.

Wide current measurement range. Measures currents up to 30A, which suits various devices.

Enhanced safety. The non-invasive design reduces risks compared to traditional invasive meters.

The prototype of the IoT smart meter includes an SSD1306 OLED display module (LCD1), as shown in Fig. 1. The components required to build the IoT smart meter are listed in the Bill of Materials table.

The circuit and working

Fig. 2 shows the circuit diagram of the IoT smart meter. It is built around an IndusBoard coin (U2), SSD1306 OLED display module (LCD1), and non-invasive AC current sensor SCT013000100A (U1). The current sensor connects to the circuit via the IndusBoard coin.

Alternatively, the design supports an OLED or a round LCD display. Fig. 3 illustrates the circuit using a GC9A01 3.25cm (1.28-inch) round LCD TFT display, substituting the SSD1306 OLED display. The remaining circuit elements remain the same.

Software

First, install the Arduino integrated development environment (IDE) and write the sketch. In the source code, the device can be configured

EFY Note

The source code of this project is available for free download at www.electronicsforu.com

to either connect to a Wi-Fi network or act as a standalone network. By default, the code is set to STA mode, allowing the device to act as its own server and network. A laptop or phone can connect to the device's Wi-Fi.

If discrepancies are observed between readings from the IoT smart meter and a standard energy meter, calibration can be performed in the source code. Fig. 4 shows a snippet of the source code.

To upload the source code:

- Plug in the IndusBoard.
- Select the ESP32S2 board in the Arduino IDE.
- Set the correct COM port and enable CDC in the tools settings.
- Upload the code using the upload option in the sketch menu.

Construction and testing

Based on the circuit diagram, the design can be converted into a printed circuit board (PCB), as shown in Fig. 5, and made or customised using the reference design. Once the PCB is ready:

- Upload the source code onto the IndusBoard coin.
- Assemble the components as shown in Fig. 2.
- If using a round display, connect the GC9A01 round display to the IndusBoard, as shown in Fig. 3.

For testing, search for the IoT